

Shortest Paths Algorithms Exercises

Lectures 5–6, Spring 2026

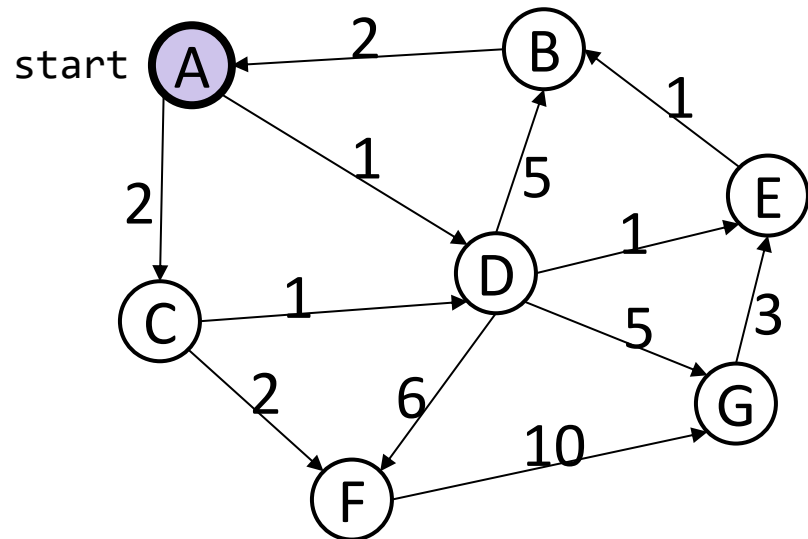
BFS

Dijkstra's Algorithm

Bellman-Ford Algorithm

Q. Dijkstra's Algorithm

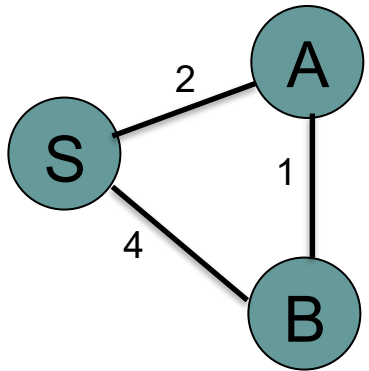
Exam question: Given this directed graph, run Dijkstra's Algo to find shortest paths starting from **source node A**. Give the node visit order, and fill in this table of SN (Shortest Distance) and PN (Previous Node), crossing out old SD and PN as you find a shortcut path with smaller SD



Visit Order

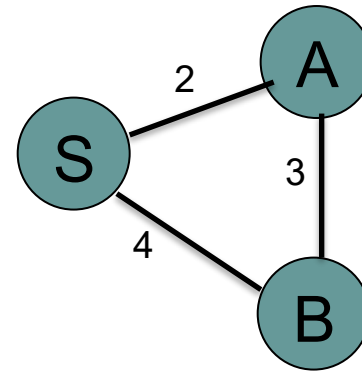
Node	SD	PN
A		
B		
C		
D		
E		
F		
G		

Q. Dijkstra's Algorithm (Source Node S)



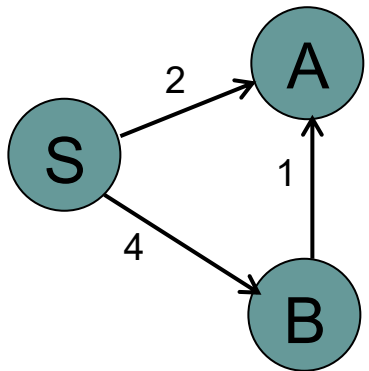
ANS

Node	SD	PN
S	0	/
A		
B		



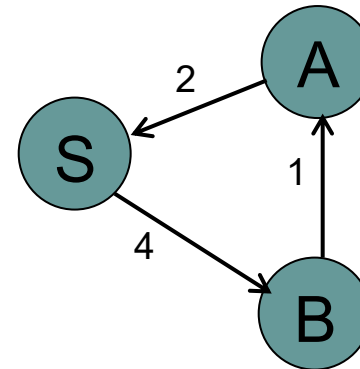
ANS

Node	SD	PN
S	0	/
A		
B		



ANS

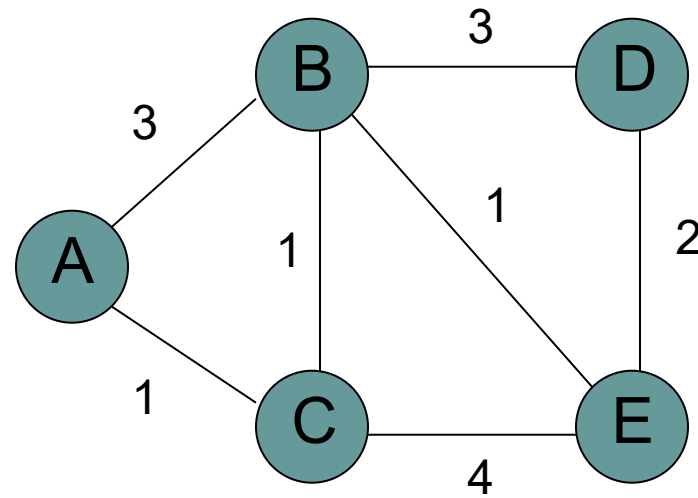
Node	SD	PN
S	0	/
A		
B		



ANS

Node	SD	PN
S	0	/
A		
B		

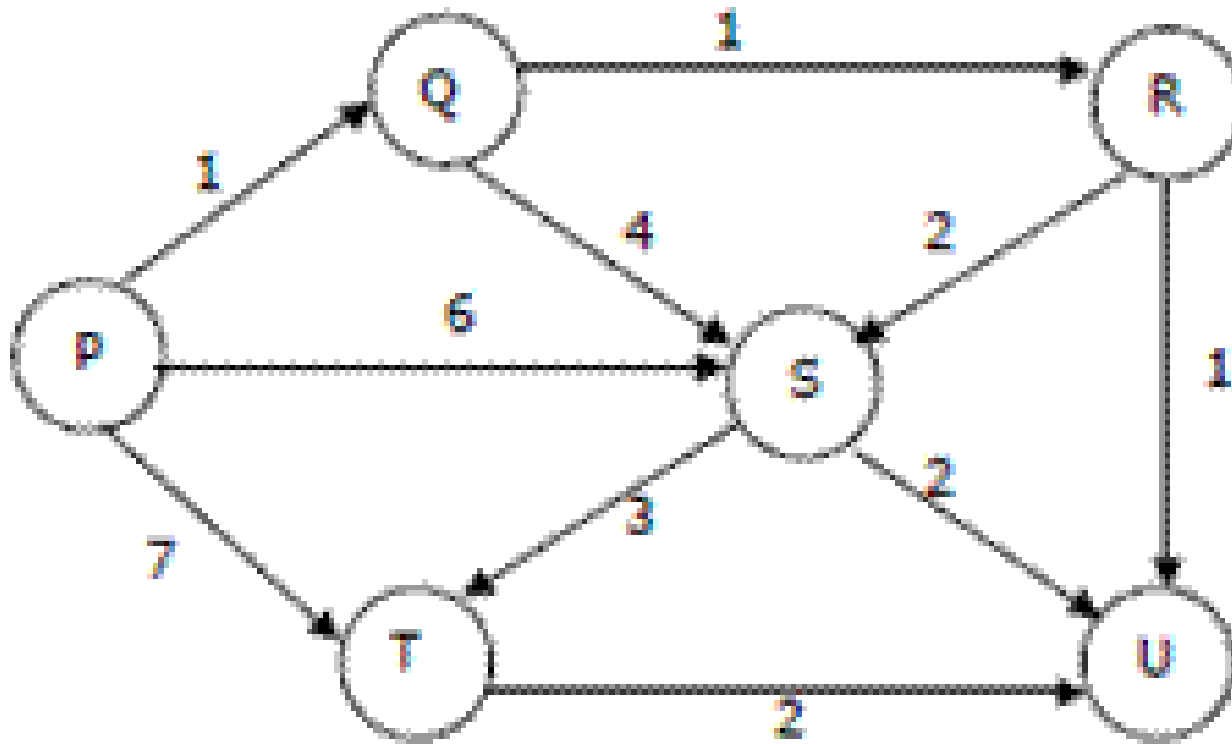
Q. Dijkstra's Algorithm (Source Node A, Undirected Graph)



Visit Order

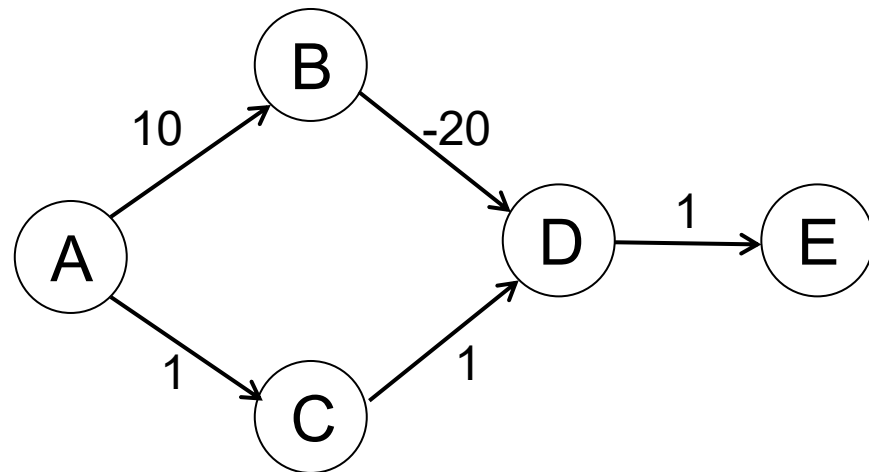
Node	SD	PN
A		
B		
C		
D		
E		

Q. Dijkstra's Algorithm (Source Node P, Directed Graph)



Q. Topological Sort

Given this directed graph, run Topological Sort to find shortest paths starting from **source node A**. Give the node visit order, and fill in this table of SN (Shortest Distance) and PN (Previous Node), crossing out old SD and PN as you find a shortcut path with smaller SD. Considering **all possible** topological orders.



Visit Order

Node	SD	PN
A	0	/
B		
C		
D		
E		

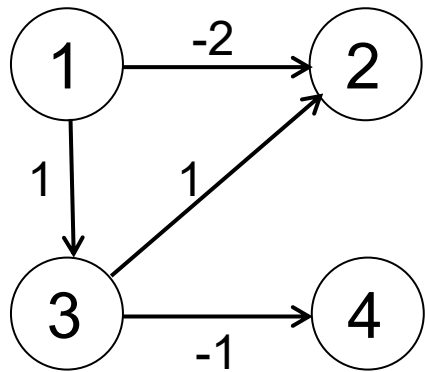
Q. Johnson's algorithm

Consider the following weighted digraph. As part of Johnson's algorithm for All-pairs Shortest Paths, add a dummy source node d , and edges with weight 0 from d to all vertices of G . Let the modified graph be G' .

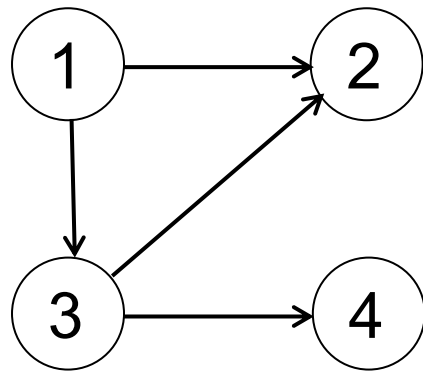
a) Compute the shortest distances from dummy source node d to each node in G' by hand: $h[0]$, $h[1]$, ..., $h[V-1]$, then reweight the edges of the original graph to make the edge weights greater than or equal to 0. Draw the reweighted graph G' (without the dummy node d).

b) For the reweighted graph G' : run Dijkstra's Algo to find shortest paths starting from **source node 1**, and compute the shortest paths for the graph with updated positive or zero weights. (Do not show the intermediate steps.)

c) For the original graph G : compute the shortest paths starting from **source node 1** with negative weights.



Original graph



Reweighted graph

Node	$h()$
1	
2	
3	
4	

Shortest paths starting from dummy node

Node	SD	PN
1	0	/
2		
3		
4		

Shortest paths starting from source node 1 in reweighted graph

Node	SD	PN
1	0	/
2		
3		
4		

Shortest paths starting from source node 1 in original graph